ADITYA UNIVERSITY

SCHOOL OF COMPUTING

Department of Computer Science and Engineering
DATA ANALYSIS ESSENTIALS

CERTIFICATE

This is to certify that the Cornerstone project work entitled *"Vehicle Fuel Analysis System"* is submitted by

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in partial fulfilment of the requirements for the award of the B.Tech. degree in Computer Science and Engineering for the academic year 2026-2027.

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DECLARATION

We hereby declare that the project entitled “VEHICLE FUEL ANALYSIS SYSTEM” is our original work and is submitted to ADITYA UNIVERSITY, Surampalem, in partial fulfillment of the requirements for the award of the B.Tech. degree in Computer Science and Engineering.

We further declare that this project work has not been submitted, either in full or in part, for the award of any degree or diploma in this or any other institution. We also confirm that this work has been carried out with academic honesty and complies with the Institutional Academic Integrity and AI Usage Policy.

Place: Surampalem

Date: 03-04-2026

By

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ABSTRACT

With the increasing cost of fuel and the growing need for efficient transportation, analysing vehicle fuel consumption has become essential for reducing operating costs and improving vehicle performance. The Vehicle Fuel Analysis System is designed to collect, process, and analyse fuel consumption data to provide meaningful insights into vehicle efficiency. The system analyses vehicle specifications and fuel-efficiency information such as vehicle model, manufacturer, engine displacement, fuel type, fuel tank capacity, and fuel efficiency. Using Python and Pandas, the data is cleaned, analysed, and visualised to identify fuel-efficiency patterns and compare vehicle performance. The results are presented through bar charts, scatter plots, and histograms. This system provides an accurate, user-friendly, and cost-effective solution for monitoring fuel consumption and generating performance reports. It helps vehicle owners and fleet managers optimise fuel usage, reduce unnecessary expenses, and improve operational efficiency. Future enhancements include GPS integration, real-time fuel monitoring using IoT sensors, predictive maintenance, and AI-based fuel consumption forecasting to support intelligent transportation management.

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CHAPTER 1 – INTRODUCTION

1.1 Introduction to the Project

The Vehicle Fuel Analysis System is a data analysis project developed to study vehicle fuel efficiency and understand the factors that influence fuel consumption. Fuel efficiency is an important aspect of vehicle performance because it affects operating costs and helps in making better vehicle comparisons.

Vehicles have different engine specifications, dimensions, fuel types, transmission systems, and efficiency values. Analysing these attributes together can help identify patterns in vehicle fuel efficiency.

In this project, the Cars India Dataset is analysed using Python and Pandas. The dataset contains information about different vehicles, including their model, maker, engine specifications, fuel type, fuel tank capacity, and fuel efficiency. The data is inspected, cleaned, and prepared for further analysis and visualization.

The project follows a simple data-analysis process:

Collect → Clean → Analyse → Visualise → Identify Insights
The analysis helps in comparing vehicles and understanding relationships between vehicle characteristics and fuel efficiency.

1.2 Problem Statement

Vehicle owners and fleet managers need a clear way to understand fuel efficiency and compare different vehicles. When many vehicle specifications are considered manually, it can be difficult to identify useful relationships and fuel-consumption patterns.

Manual analysis of vehicle data can make it difficult to:

- Track and compare vehicle fuel efficiency.
- Identify fuel-consumption trends.
- Understand the relationship between engine characteristics and efficiency.
- Compare vehicles based on fuel type and other specifications.
- Identify unusual values in the dataset.

Therefore, there is a need for a data-driven system that can analyse and visualize vehicle data and provide meaningful insights into fuel-efficiency patterns.

1.3 Objectives of the Project

The main objectives of the Vehicle Fuel Analysis System are:

1. To analyse vehicle specifications and fuel-efficiency values.

2. To compare vehicles using fuel type, engine specifications, and efficiency attributes.

3. To perform data-quality analysis and identify missing values in the dataset.

4. To study relationships between vehicle characteristics and fuel efficiency.

5. To visualize the analysed data using meaningful charts and comparisons.

6. To identify useful patterns in vehicle fuel efficiency through data analysis.

1.4 Scope of the Project

The scope of the Vehicle Fuel Analysis System is to analyse vehicle-related data and study fuel-efficiency patterns using data-analysis techniques.

The project focuses on:

- Analysing vehicle and engine specifications.
- Studying different fuel types.
- Comparing fuel-efficiency values of vehicles.
- Cleaning and preparing the dataset for analysis.
- Identifying missing and duplicate data.
- Studying relationships between vehicle characteristics and fuel efficiency.
- Presenting analysis results through graphs and visualizations.
- Providing data-based insights for understanding vehicle fuel efficiency.

The current project is based on the available Cars India Dataset. Future extensions mentioned in the project include GPS integration, real-time fuel monitoring using IoT sensors, predictive maintenance, and AI-based fuel-consumption forecasting.

CHAPTER 2 – DATASET DESCRIPTION AND UNDERSTANDING

2.1 Dataset Source

The dataset used in this project is the Cars India Dataset, provided in CSV format.

S.No	Property	Details
1	Dataset Name	Cars India Dataset
2	File Format	CSV
3	Number of Records	156
4	Number of Attributes	24
5	Data Types	Numerical and Categorical

Table.1: Dataset Details

The dataset was selected because it is directly related to the vehicle fuel-efficiency analysis project. It contains vehicle, engine, fuel, efficiency, safety, and other vehicle-related information, and its mix of numerical and categorical attributes makes it suitable for data cleaning, preprocessing, comparison, and visualization.

2.2 Dataset Description

Each record represents vehicle-related information, while the columns represent different attributes of the vehicles. The dataset includes information related to:

- Vehicle identity
- Engine specifications
- Fuel
- Fuel efficiency
- Vehicle dimensions
- Transmission
- Braking system
- Drive configuration
- Safety
- Other vehicle specifications

2.3 Features / Attributes

The following are the 24 attributes available in the Cars India Dataset:

S.No.	Attribute	Description
1	Model	Car model name
2	Maker	Vehicle manufacturer
3	Type	Vehicle/body type
4	Seats	Number of seats
5	Displacement	Engine displacement
6	Length	Vehicle length
7	Width	Vehicle width
8	Height	Vehicle height
9	Wheelbase	Distance between front and rear axles
10	No_of_Cylinders	Number of engine cylinders
11	Fuel	Fuel type
12	Engine Type	Type of engine
13	Transmission	Transmission type
14	Front Brake	Front braking system
15	Rear Brake	Rear braking system
16	Drive	Drive configuration
17	Turning Radius	Minimum turning radius
18	Fuel Tank Capacity	Fuel tank capacity
19	Boot Space	Boot/luggage capacity
20	Fuel Efficiency	Fuel efficiency / mileage
21	Emission Type	Emission standard/type
22	Tyre Size	Tyre specification
23	Variants	Number of variants
24	NCAP Rating	NCAP safety rating

Table 2: Cars India Dataset – 24 Attributes

Important features for the project: Model, Maker, Type, Displacement, No. of Cylinders, Fuel, Fuel Tank Capacity, Transmission, and Fuel Efficiency.

2.4 Target Variable, if Applicable

The project is primarily focused on data analysis and visualization, rather than developing a predictive machine-learning model. Therefore, a formal machine-learning target variable is not defined at this stage.

However, Fuel Efficiency is the main analysis variable because the project aims to study fuel-efficiency patterns and compare vehicles based on their efficiency values.

2.5 Dataset Statistics / Initial Exploration

Initial exploration was performed using the following Pandas functions:

```
import pandas as pd
df = pd.read_csv("Cars India Dataset.csv")
df.head()
df.shape
df.info()
```

S.No.	Parameter	Result
1	Records	156
2	Attributes	24
3	Missing Cells	62
4	Duplicate Rows	0

Table.3: Initial Dataset Summary

The initial check identified 62 missing cells across 8 columns, while no duplicate rows were found. The main missing-value concentrations were in Fuel Efficiency, Turning Radius, and Boot Space.

CHAPTER 3 – DATA PREPROCESSING

3.1 Data Collection / Loading

The dataset is loaded into Python using the Pandas library and stored in a DataFrame named df:

```
import pandas as pd
df = pd.read_csv("Cars India Dataset.csv")
```

Data Loading Flow: Cars India Dataset → CSV File → Python → Pandas DataFrame → Initial Inspection

3.2 Handling Missing Values

The initial inspection showed 62 missing cells across 8 columns. To handle missing values:

- Numerical columns are treated using the median value.
- Categorical columns are treated using the mode value.

After the missing-value treatment, the reported result was: Missing values = 0.

3.3 Handling Duplicate / Inconsistent Data

The initial dataset contained 0 duplicate rows, so no duplicate records needed to be removed. Data types were inspected to distinguish numerical from categorical data, since different types require appropriate preprocessing methods.

Preprocessing process: Raw Data → Check → Clean → Transform → Analysis-Ready Data

3.4 Feature Scaling

Feature scaling was not applied. The project focuses on descriptive data analysis and visualization, and numerical attributes such as Displacement and Fuel Efficiency are retained in their original units so that results remain directly interpretable.

3.5 Standardization

Statistical standardization was not implemented, because the project performs descriptive analysis, comparison, aggregation, and visualization rather than training a predictive model.

3.6 Label Encoding

Label encoding was not applied. Categorical attributes such as Maker, Fuel, Type, Engine Type, and Transmission are used directly for grouping, filtering, comparison, and descriptive analysis.

3.7 One-Hot Encoding

One-hot encoding was not used for the analysis pipeline, because the project does not train a supervised machine-learning model. (A pd.get_dummies() experiment produced a 193-column encoded DataFrame that was not used in later analysis, which works on the 24-column cleaned dataset.)

3.8 Other Project-Specific Preprocessing

After missing-value treatment and duplicate checking, the cleaned DataFrame is exported as a separate CSV file, which becomes the input for the main analysis stage:

```
df_clean.to_csv("Cars_India_Cleaned.csv", index=False)
```

CHAPTER 4 – SYSTEM DESIGN AND METHODOLOGY

4.1 Proposed Methodology

1. Load the Cars India dataset into a Pandas DataFrame.

2. Inspect the dataset structure, dimensions, data types, missing values, and duplicate records.

3. Treat the identified missing values according to the data type.

4. Verify duplicate records and prepare the analysis-ready dataset.

5. Export the cleaned DataFrame as Cars_India_Cleaned.csv.

6. Read the cleaned dataset for the main analysis stage.

7. Compare Fuel Efficiency values across vehicle models.

8. Perform manufacturer-wise analysis using Maker and average Fuel Efficiency.

9. Study the relationship between Displacement and Fuel Efficiency.

10. Compare fuel-efficiency patterns across Fuel categories where applicable.

11. Generate graphical visualizations using Matplotlib.

12. Interpret the analytical outputs and summarize the findings.

4.2 System Workflow / Architecture

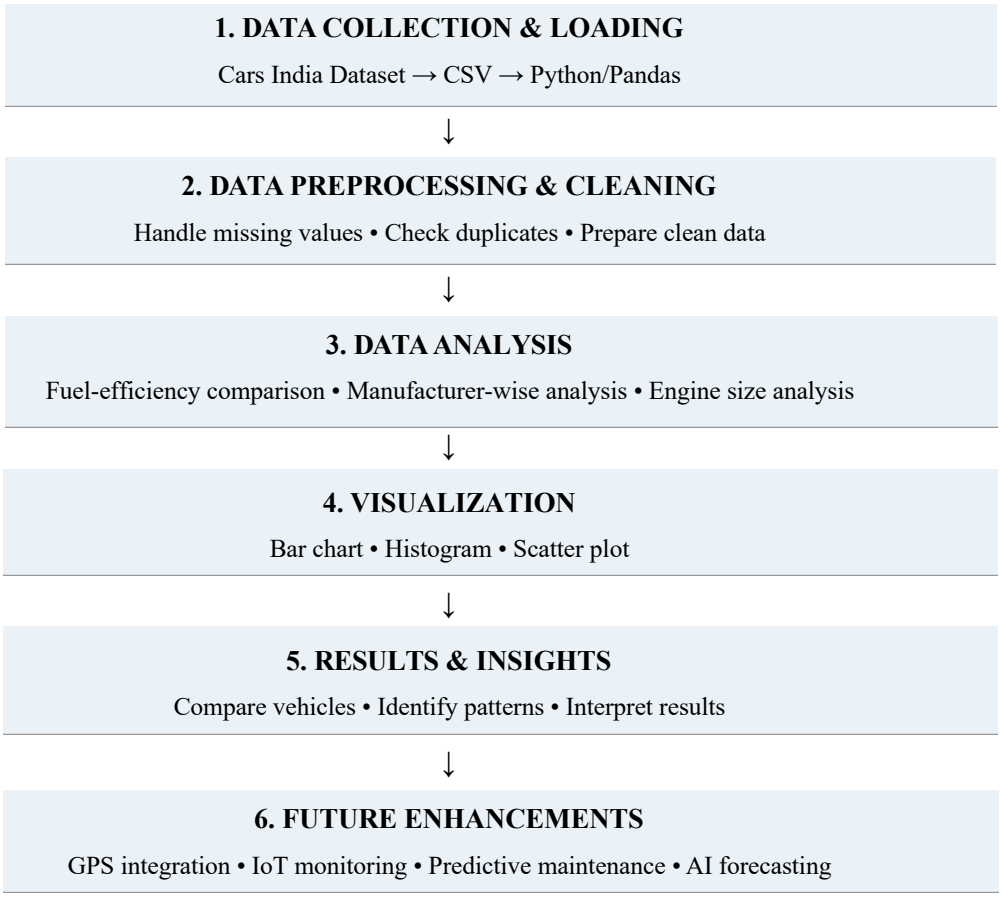
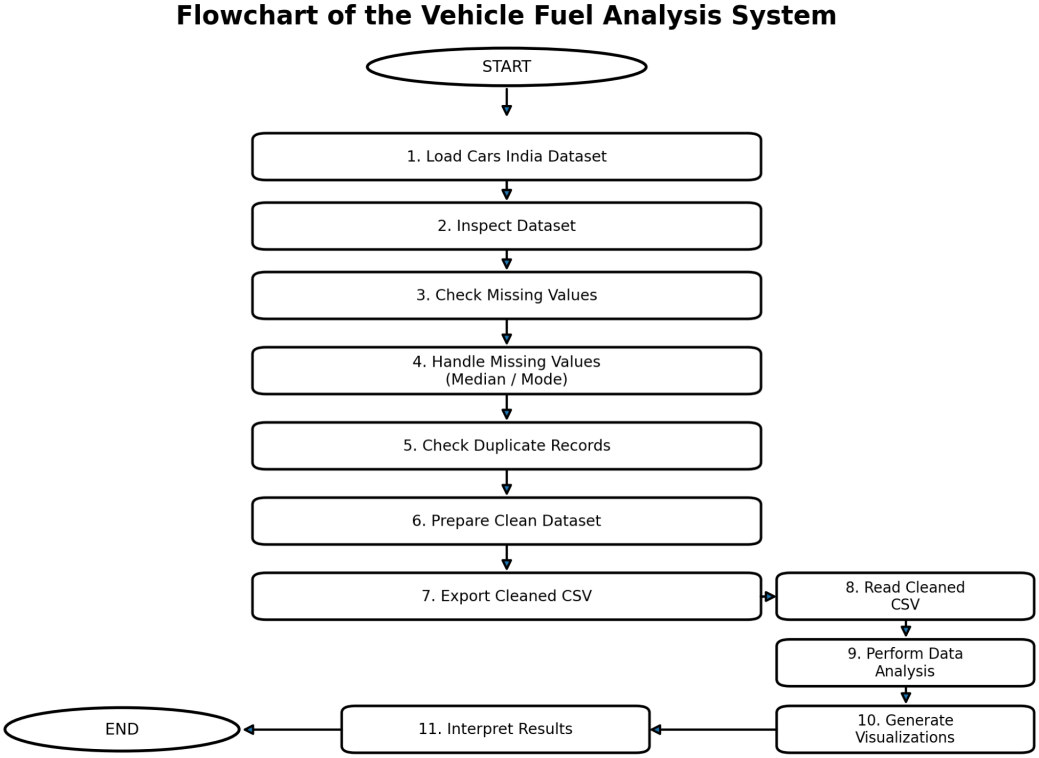


Fig. 1. Project Workflow of the Vehicle Fuel Analysis System

4.3 Flowchart



4.4 Tools and Technologies Used

S.No.	Tool / Technology	Purpose
1	Python	Programming language used for data analysis
2	Pandas	Data loading, cleaning, grouping and analysis
3	NumPy	Numerical operations and preprocessing
4	Matplotlib	Data visualization and chart generation
5	CSV	Dataset input and cleaned-data storage format
6	Google Colab / Jupyter Notebook	Development and execution environment

Table 4: Tools and Technologies Used

4.5 Algorithm / Model Used

The Vehicle Fuel Analysis System is a descriptive data-analysis project and does not implement a supervised machine-learning prediction model. The main analytical operations are:

- Descriptive analysis to understand the structure and values in the vehicle dataset.
- Fuel-efficiency comparison by selecting Model, Maker, and Fuel Efficiency and sorting the results.
- Manufacturer-wise aggregation by grouping vehicles using Maker and calculating mean Fuel Efficiency.
- Engine-size versus mileage analysis using Displacement and Fuel Efficiency.
- Correlation analysis between Displacement and Fuel Efficiency.
- Graphical visualization using Matplotlib.

CHAPTER 5 – IMPLEMENTATION

5.1 Development Environment

The project was implemented in a Google Colab notebook using Python. The libraries used are Pandas, NumPy, and Matplotlib. Google Drive was mounted in the notebook so that the dataset file could be read from Drive. The input is Cars India Dataset.csv, and the cleaned output is Cars_India_Cleaned.csv.

5.2 Dataset Loading and Dataset Exploration

Dataset Loading

```
from google.colab import drive
drive.mount('/content/drive')
import pandas as pd
df = pd.read_csv("Cars India Dataset.csv")
```

Exploration

```
df.head()              # first five rows
df.shape               # (156, 24)
df.info()              # columns, data types, non-null counts
df.isnull().sum()      # missing values per column
df.isnull().sum().sum()  # output: 62
df.select_dtypes(include='number').columns
df.duplicated().sum()    # output: 0
```

The numerical columns are: Seats, Displacement, Length, Width, Height, Wheelbase, No_of_Cylinders, Turning Radius, Fuel Tank Capacity, Boot Space, Fuel Efficiency, and Variants. The duplicate check returns 0.

5.3 Data Preprocessing

Missing value treatment (median for numerical, mode for categorical):

```
import numpy as np
numeric_columns = df.select_dtypes(include=np.number).columns
categorical_columns = df.select_dtypes(include='object').columns
for col in numeric_columns:
    df[col] = df[col].fillna(df[col].median())
for col in categorical_columns:
    df[col] = df[col].fillna(df[col].mode()[0])
df.isnull().sum().sum()   # output: 0
```

Duplicate removal:

```
df.duplicated().sum()              # output: 0
df.drop_duplicates(inplace=True)
print("Duplicates after removal:", df.duplicated().sum())
```

Categorical encoding (experiment, not used in final analysis):

```
df_encoded = pd.get_dummies(df, drop_first=True)
df_encoded.head()          # 193 columns
```

Outlier handling using the IQR method:

```
df[numeric_columns].describe()
df_clean = df.copy()
for col in numeric_columns:
    Q1 = df_clean[col].quantile(0.25)
    Q3 = df_clean[col].quantile(0.75)
    IQR = Q3 - Q1
```

```
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    df_clean = df_clean[
        (df_clean[col] >= lower) &
        (df_clean[col] <= upper)
    ]
```

Final check:

```
print("Final dataset shape:", df_clean.shape)   # (93, 24)
print("Missing values:", df_clean.isnull().sum().sum())      # 0
print("Duplicate rows:", df_clean.duplicated().sum().sum())  # 0
display(df_clean.head())
```

Cleaned Dataset Export

```
df_clean.to_csv("Cars_India_Cleaned.csv", index=False)
print("Cleaned dataset saved successfully!")
import pandas as pd
df_clean = pd.read_csv("Cars_India_Cleaned.csv")
print(df_clean.shape)   # (93, 24)
df_clean.head()
```

5.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed on the cleaned Cars India dataset to understand the structure, characteristics, and important patterns in the vehicle data before performing the main project analysis.

The main EDA activities performed were:

1.Dataset Structure Analysis

The cleaned dataset was examined to understand the number of records, attributes, data types, and sample records.

2. Statistical Summary

Descriptive statistics were generated for numerical attributes such as Displacement, Fuel Tank Capacity, and Fuel Efficiency to understand their minimum, maximum, average, and spread.

3. Fuel Efficiency Analysis

The Fuel Efficiency attribute was examined to understand its range and general distribution across the available vehicles.

4. Categorical Attribute Analysis

Categorical attributes such as Maker and Fuel were examined to understand the number of vehicles in different manufacturer and fuel categories.

5. Data Quality Verification

The cleaned dataset was checked again for missing values and duplicate records to ensure that the data was suitable for further analysis.

6.Relationship Exploration

The Displacement and Fuel Efficiency attributes were examined as important variables for understanding the relationship between engine size and vehicle efficiency.

EDA helped in identifying the important attributes required for the detailed fuel-efficiency comparison, manufacturer-wise analysis, and engine-size analysis performed in the next stage.

5.5 Model / Analysis Implementation

5.5.1 Fuel Efficiency Comparison

```
fuel_comparison = df_clean[
    ["Model", "Maker", "Fuel Efficiency"]
].sort_values(
    by="Fuel Efficiency",
    ascending=False
)
print(fuel_comparison)
```

The output lists vehicles from the highest to the lowest fuel efficiency, which makes the variation in mileage easy to see.

5.5.2 Manufacturer-wise Analysis

```
manufacturer_analysis = df_clean.groupby("Maker")[
    "Fuel Efficiency"
].mean().sort_values(ascending=False)
print(manufacturer_analysis)
```

S.No.	Maker	Average Fuel Efficiency (km/L)
1	Toyota	20.27
2	Honda	19.73
3	Hyundai	19.16
4	Volkswagen	18.67
5	Mahindra	17.57
6	Kia	17.49
7	Nissan	15.30

Table 5: Average Fuel Efficiency by Maker (Cleaned Dataset)

5.5. Engine Size vs Fuel Efficience

```
# engine displacement vs mileage
engine_mileage = df_clean[
   ["Model", "Displacement", "Fuel Efficiency"]
].sort_values(
   by="Displacement"
)
print(engine_mileage)

# correlation
correlation = df_clean[["Displacement", "Fuel Efficiency"]].corr()
print(correlation)

correlation = df_clean["Displacement"].corr(df_clean["Fuel Efficiency"])
print("Correlation:", correlation)
```

The correlation coefficient lies between −1 and +1. The result for this dataset is **−0.3389** (approximately −0.339): vehicles with larger displacement tend to have somewhat lower fuel efficiency, but the relationship is not strong — displacement alone does not determine fuel efficiency, because other attributes also affect it.

5.6 Code Explanation

Code Section	What it does	Why it is used	Output
pd.read_csv()	Loads the CSV file into a DataFrame	To start the analysis in Python	DataFrame of 156 × 24
isnull().sum()	Counts missing values	To check data quality	62 missing cells
fillna() with median / mode	Fills missing values	To remove empty values	0 missing values
duplicated()	Counts repeated rows	To avoid repeated records	0 duplicates
IQR loop	Removes rows outside the IQR bounds	To handle outliers	93 × 24 dataset
to_csv()	Saves the cleaned data	To reuse the clean data	Cleaned CSV file
sort_values()	Sorts vehicles by Fuel Efficiency	To compare vehicle models	Sorted list of models
groupby().mean()	Averages Fuel Efficiency by Maker	To compare manufacturers	7 maker averages
corr()	Calculates correlation	To study displacement vs efficiency	−0.3389
plt.scatter / bar / hist	Draws the graphs	To show results visually	Three charts

Table 6: Summary of Important Code Sections

CHAPTER 6 – DATA VISUALIZATION AND RESULTS

After analysing the cleaned vehicle dataset, the results are visualized using Python and Matplotlib.

Visualization	Chart Type	Purpose	Data Used
Manufacturer-wise Fuel Efficiency	Bar Chart	Compares average fuel efficiency of manufacturers	Maker + Average Fuel Efficiency
Fuel Efficiency Distribution	Histogram	Shows the distribution of fuel-efficiency values	Fuel Efficiency
Engine Size vs Mileage	Scatter Plot	Shows the relationship between displacement and fuel efficiency	Displacement + Fuel Efficiency

Table 7: Data Visualization Summary

6.1 Manufacturer-wise Fuel Efficiency – Bar Chart

The manufacturer-wise bar chart compares the average Fuel Efficiency of different manufacturers. The Maker attribute is used to group vehicles, and the mean Fuel Efficiency is calculated for each group.

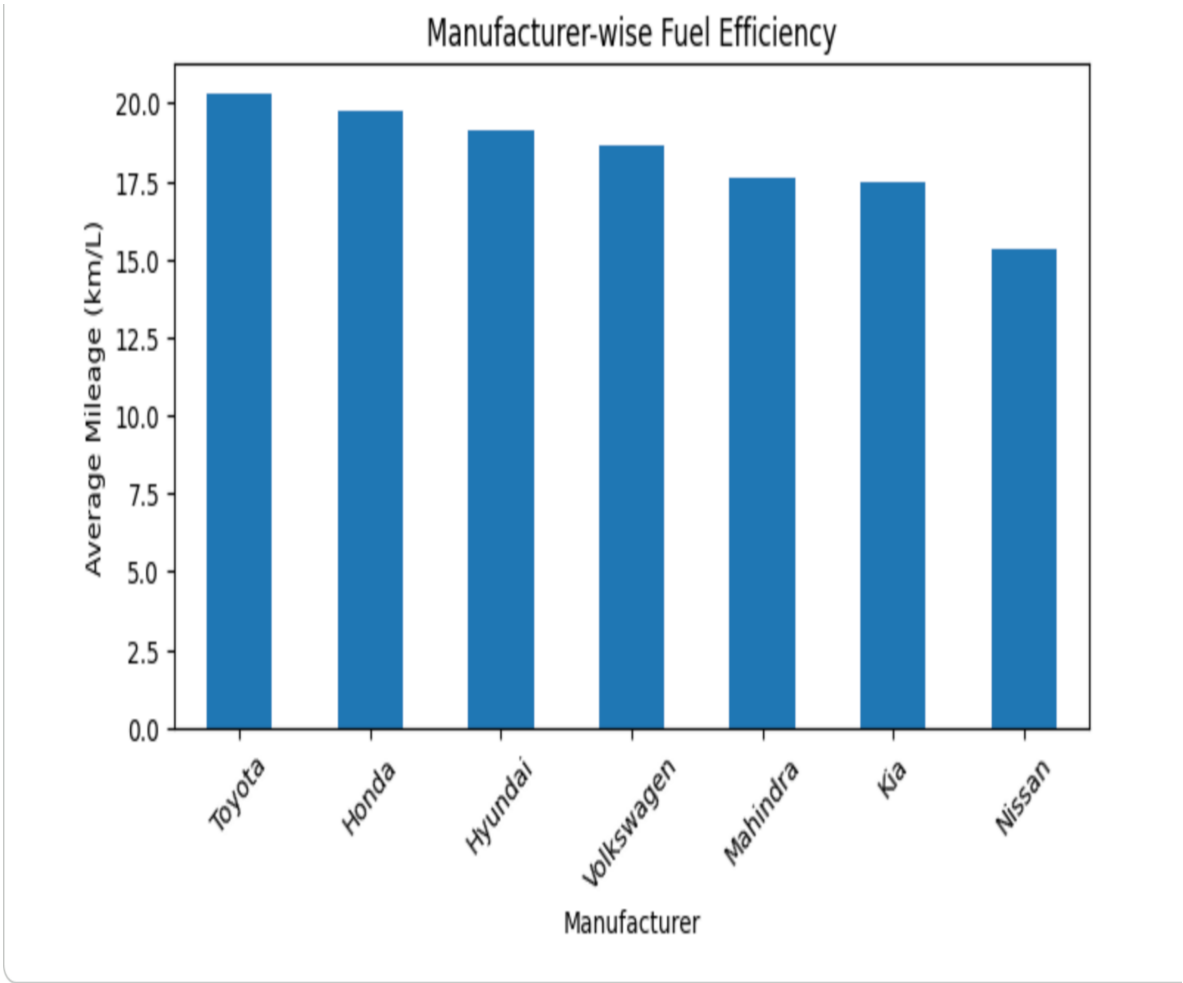


Fig. 3. Manufacturer-wise Fuel Efficiency Bar Chart

6.2 Fuel Efficiency Distribution – Histogram

The histogram shows how Fuel Efficiency values are distributed across the vehicles in the dataset. It helps identify the concentration and spread of mileage values.

- **X-axis:** Fuel Efficiency (km/L)
- **Y-axis:** Number of Vehicles

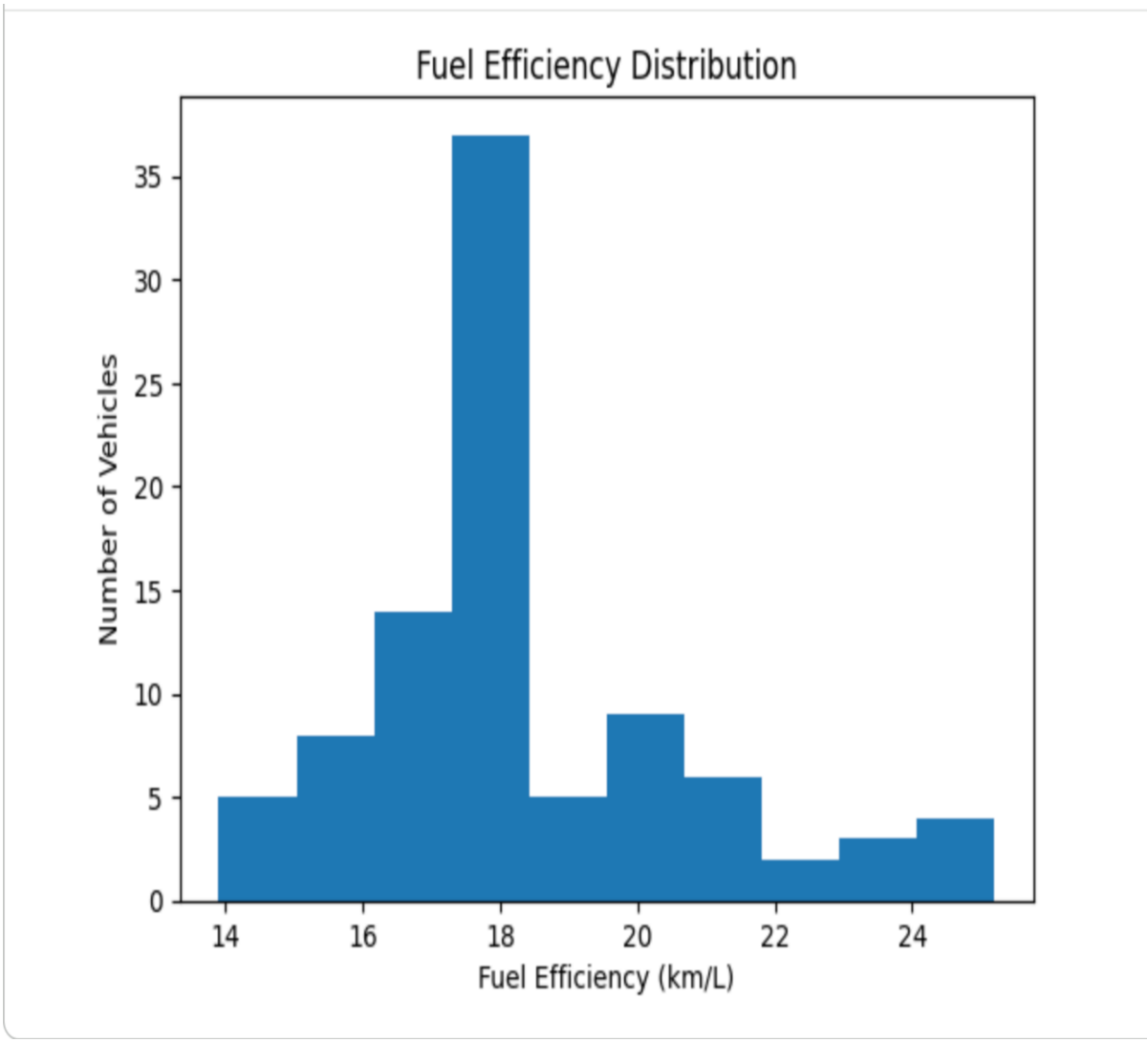


Fig. 4. Fuel Efficiency Distribution Histogram

6.3 Engine Size vs Mileage – Scatter Plot

The scatter plot compares engine displacement with Fuel Efficiency. Each point corresponds to a vehicle.

- **X-axis:** Engine Size (Displacement)
- **Y-axis:** Mileage (Fuel Efficiency)

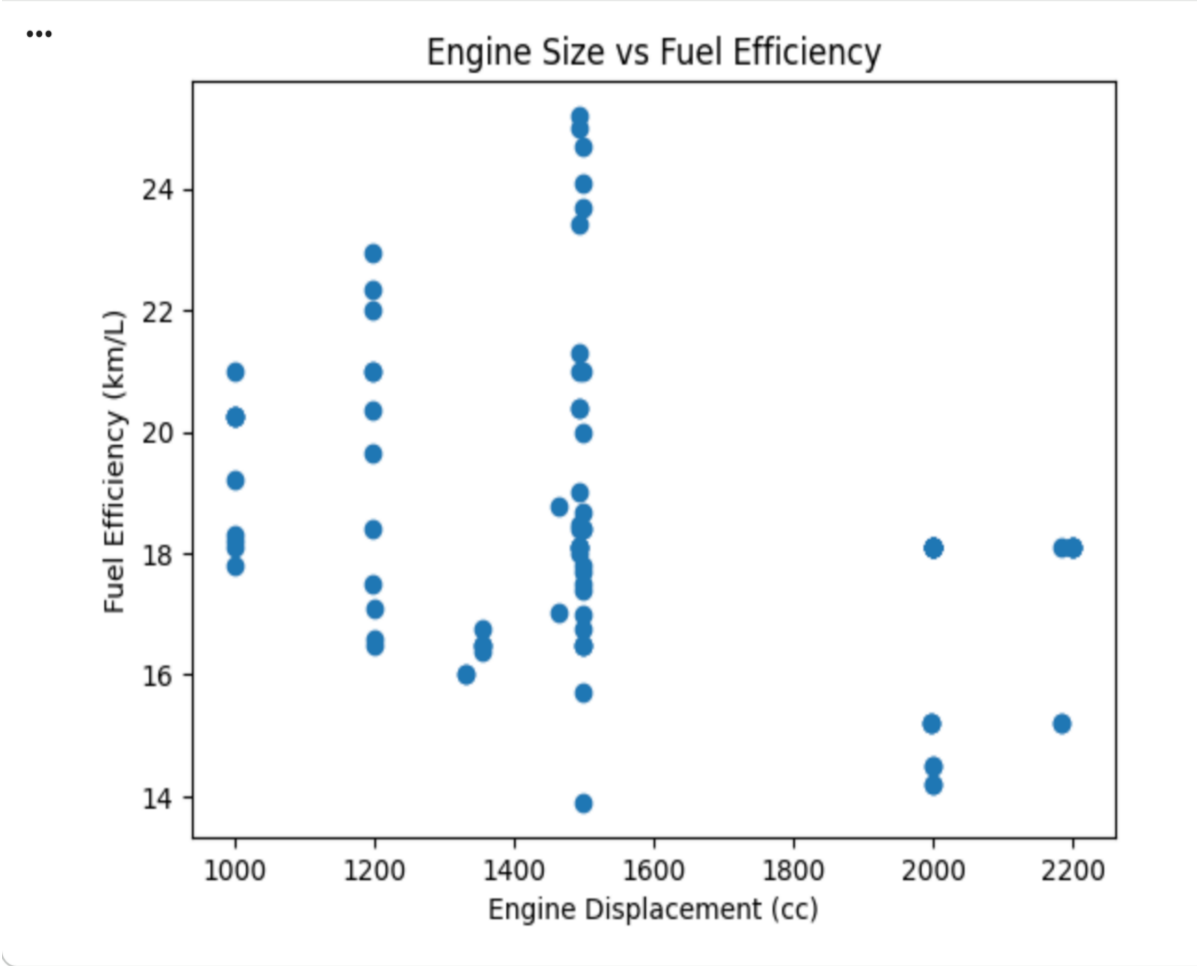


Fig. 5.Engine Size vs Mileage Scatter Plot

6.4 Project-Specific Visualizations

Fuel Efficiency Comparison

Fuel-efficiency comparison is performed at the individual vehicle level by sorting Fuel Efficiency values. This output supports direct comparison of vehicle models and complements the manufacturer-level analysis.

	Model	Maker	Fuel Efficiency
53	i20	Hyundai	25.2
58	Verna	Hyundai	25.0
1	Amaze	Honda	24.7
5	City 5th Gen	Honda	24.1
8	WR-V	Honda	23.7
..	...	...	...
70	Alcazar	Hyundai	14.5
74	Alcazar	Hyundai	14.5
75	Alcazar	Hyundai	14.2
71	Alcazar	Hyundai	14.2
13	Kicks	Nissan	13.9

[93 rows x 3 columns]

Fig. 6.Vehicle-wise Fuel Efficiency Comparison

6.5 Interpretation of Visualizations

The manufacturer-wise bar chart provides a direct comparison of average Fuel Efficiency across manufacturers. The histogram shows the distribution of efficiency values across the available vehicles. The scatter plot provides a visual representation of the association between engine displacement and Fuel Efficiency.

The project uses visualization for comparison and pattern identification. The graphs should be interpreted within the limits of the available dataset; they do not by themselves establish that one vehicle characteristic causes another.

CHAPTER 7 – ANALYSIS AND DISCUSSION OF RESULTS

7.1 Evaluation Metrics

The project is evaluated using the following analysis measures:

- **Dataset Size:** The original dataset contains 156 records and 24 attributes.
- **Missing Value Check:** The original dataset contained 62 missing cells. After preprocessing, the cleaned dataset contains 0 missing values.
- **Duplicate Check:** The original dataset contained 0 duplicate rows, and the cleaned dataset also contains 0 duplicate rows.
- **Descriptive Statistics:** Mean, minimum, maximum, and other statistical measures are used to understand numerical attributes.
- **Manufacturer-wise Average:** Mean Fuel Efficiency is calculated for each manufacturer.
- **Correlation Analysis:** The correlation between Displacement and Fuel Efficiency is calculated to measure their linear association.

These measures are used to evaluate the quality of the data and understand the results produced by the analysis.

7.2 Experimental Results

The implemented analysis was performed on the cleaned Cars India dataset.

The original dataset contained 156 records and 24 attributes. During preprocessing, 62 missing cells were identified and treated using median values for numerical attributes and mode values for categorical attributes. No duplicate rows were found in the original dataset. After preprocessing and IQR-based outlier handling, the final cleaned dataset contained 93 records and 24 attributes, with 0 missing values and 0 duplicate rows.

The main analysis produced the following results:

Fuel Efficiency Comparison:

Vehicle records were sorted according to Fuel Efficiency to allow comparison of individual vehicle models.

Manufacturer-wise Analysis:

Average Fuel Efficiency was calculated for each manufacturer. The documented averages include Toyota – 20.27 km/L, Honda – 19.73 km/L, Hyundai – 19.16 km/L, Volkswagen – 18.67 km/L, Mahindra – 17.57 km/L, Kia – 17.49 km/L, and Nissan – 15.30 km/L.

Engine Size vs Fuel Efficiency:

The relationship between Displacement and Fuel Efficiency was analysed using correlation. The correlation coefficient obtained was approximately −0.3389.

Visualization Results:

The analysis results were represented using a manufacturer-wise bar chart, fuel-efficiency histogram, and engine-size versus fuel-efficiency scatter plot.

7.3 Result Interpretation

The results obtained from the Vehicle Fuel Analysis System provide a descriptive understanding of fuel-efficiency patterns in the Cars India Dataset.

The fuel-efficiency comparison helps examine differences among individual vehicle models based on their Fuel Efficiency values.

The manufacturer-wise analysis shows that average Fuel Efficiency varies across manufacturers in the cleaned dataset. This provides a group-level comparison of fuel-efficiency values. Pasted text

The engine-size analysis shows a correlation value of approximately −0.3389 between Displacement and Fuel Efficiency. This indicates a negative association, meaning vehicles with larger displacement tend to have somewhat lower fuel efficiency in this dataset. However, the relationship is not strong, so Displacement alone cannot be considered sufficient to determine Fuel Efficiency. Pasted text

The bar chart makes manufacturer-wise differences easy to compare, the histogram shows the distribution of fuel-efficiency values, and the scatter plot visually represents the association between engine displacement and fuel efficiency. These results are descriptive and are limited to the vehicles represented in the dataset.

7.4 Discussion

The combined analysis demonstrates how a cleaned vehicle dataset can be used to study fuel-efficiency patterns from multiple perspectives:

- Individual vehicle comparison provides model-level information.
- Manufacturer-wise aggregation provides group-level comparison.
- Engine-size analysis examines the relationship between an important engine attribute and mileage.

The visualizations support the numerical analysis by presenting the results in graphical form. The findings are descriptive and depend on the vehicles represented in the Cars India Dataset.

CHAPTER 8 – OUTPUT SCREENSHOTS

8.1 Input / Dataset Screenshots

```
df_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 93 entries, 0 to 92
Data columns (total 24 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Model               93 non-null     object
 1   Maker               93 non-null     object
 2   Type                93 non-null     object
 3   Seats               93 non-null     int64
 4   Displacement        93 non-null     float64
 5   Length              93 non-null     int64
 6   Width               93 non-null     int64
 7   Height              93 non-null     int64
 8   Wheelbase           93 non-null     int64
 9   No_of_Cylinders     93 non-null     float64
 10  Fuel                93 non-null     object
 11  Engine Type         93 non-null     object
 12  Transmission        93 non-null     object
 13  Front Brake         93 non-null     object
 14  Rear Brake          93 non-null     object
 15  Drive               93 non-null     object
 16  Turning Radius      93 non-null     float64
 17  Fuel Tank Capacity  93 non-null     float64
 18  Boot Space          93 non-null     float64
 19  Fuel Efficiency     93 non-null     float64
 20  Emission Type       93 non-null     object
 21  Tyre Size           93 non-null     object
 22  Variants            93 non-null     int64
 23  NCAP Rating         93 non-null     object
dtypes: float64(6), int64(6), object(12)
memory usage: 17.6+ KB
```

Fig. 7. Dataset Information and Initial Exploration

8.2 Preprocessing Outputs

- Duplicate Verification Output; Cleaned Dataset Export Output

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.drop_duplicates(inplace=True)
```

```
print("Duplicates after removal:", df.duplicated().sum())
```

```
Duplicates after removal: 0
```

Figure 8.Duplicate Verification Output

```
df_clean.to_csv("Cars_India_Cleaned.csv", index=False)

print("Cleaned dataset saved successfully!")
```

```
Cleaned dataset saved successfully!
```

Fig. 9. Cleaned Dataset Export Output

8.3 Visualization Outputs

- Manufacturer-wise Fuel Efficiency Bar Chart; Fuel Efficiency Distribution Histogram; Engine Size vs Mileage Scatter Plot

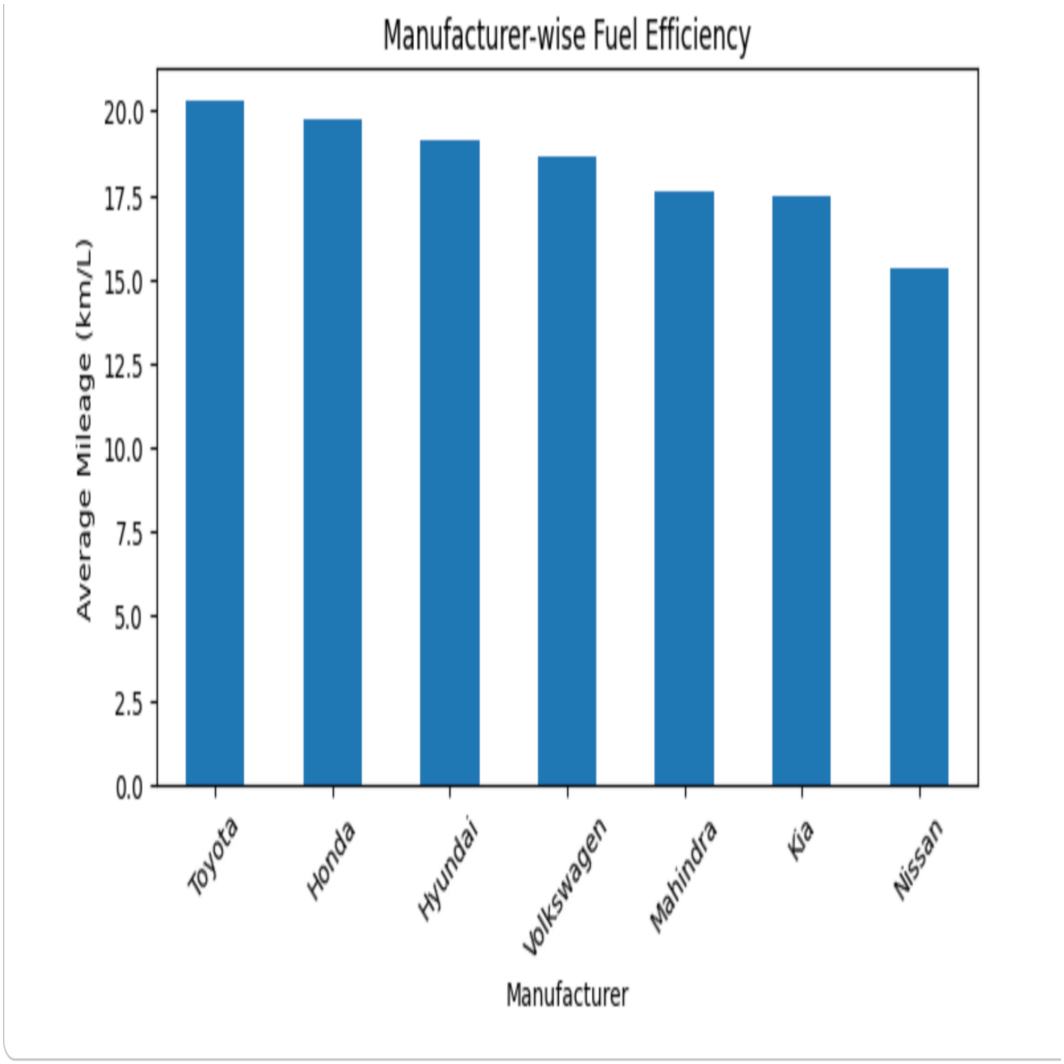


Fig. 10. Manufacturer-wise Fuel Efficiency Bar Chart

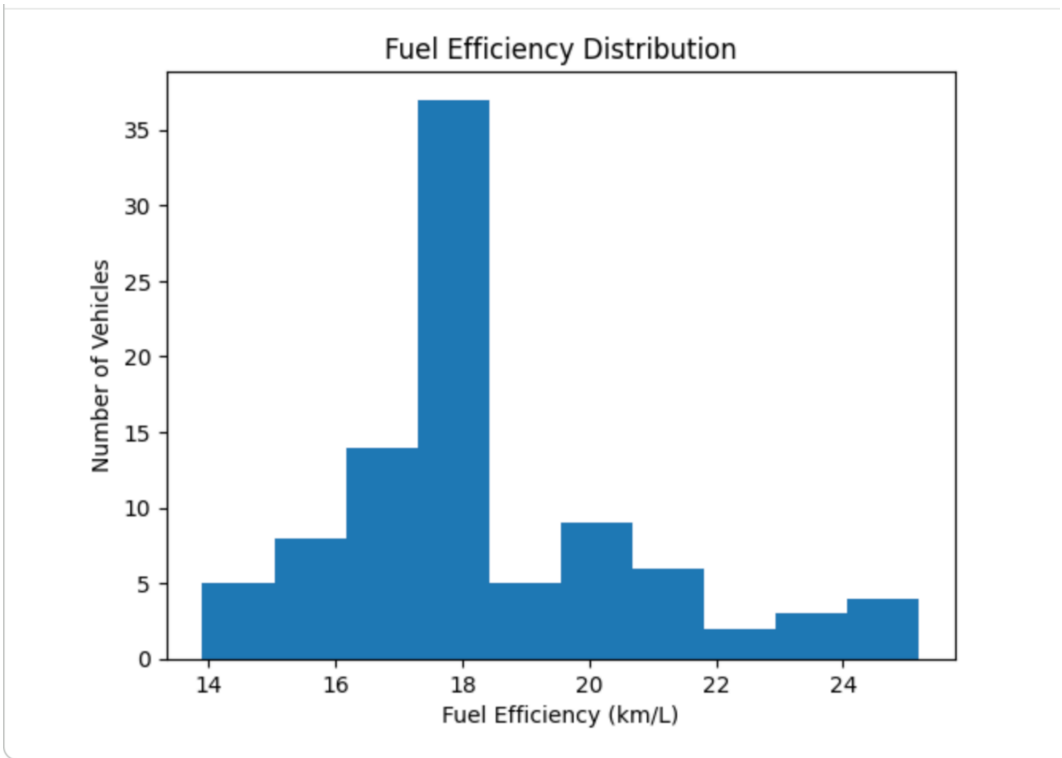


Fig. 11. Fuel Efficiency Distribution Histogram

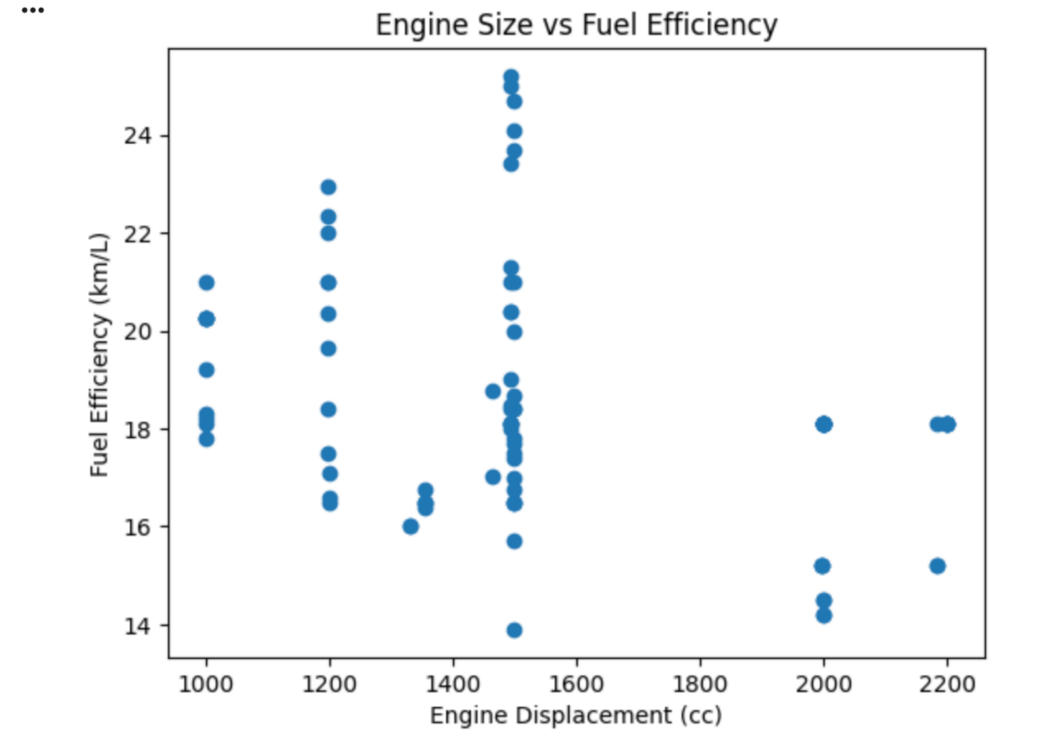


Fig. 12. Engine Size vs Mileage Scatter Plot

8.4 Prediction / Analysis Outputs

Vehicle Fuel Efficiency Comparison; Manufacturer-wise Average Fuel Efficiency; Engine Size and Mileage Analysis; Correlation Analysis Output

```
...              Model    Maker  Fuel Efficiency
    53             i20  Hyundai             25.2
    58           Verna  Hyundai             25.0
    1            Amaze    Honda             24.7
    5   City 5th Gen    Honda             24.1
    8            WR-V    Honda             23.7
    ..            ...      ...              ...
    70        Alcazar  Hyundai             14.5
    74        Alcazar  Hyundai             14.5
    75        Alcazar  Hyundai             14.2
    71        Alcazar  Hyundai             14.2
    13          Kicks   Nissan             13.9

    [93 rows x 3 columns]
```

Fig. 13. Vehicle Fuel Efficiency Comparison

```
...  Maker
    Toyota        20.270000
    Honda         19.730000
    Hyundai       19.156970
    Volkswagen    18.670000
    Mahindra      17.570370
    Kia           17.493333
    Nissan        15.300000
    Name: Fuel Efficiency, dtype: float64
```

Fig.14. Manufacturer-wise Average Fuel Efficiency

```
...                 Model  Displacement  Fuel Efficiency
    55      i20 N-Line          998.0            20.25
    48  Grand i10 Nios          998.0            21.00
    54      i20 N-Line          998.0            20.25
    51             i20          998.0            20.25
    60           Verna          998.0            19.20
    ..             ...            ...              ...
    32         XUV 700         2200.0            18.10
    33         XUV 700         2200.0            18.10
    34         XUV 700         2200.0            18.10
    35         XUV 700         2200.0            18.10
    31         XUV 700         2200.0            18.10

    [93 rows x 3 columns]
```

Fig.15.. Engine Size and Mileage Analysis

```
#correlation
correlation = df_clean[
    ["Displacement", "Fuel Efficiency"]
].corr()

print(correlation)
```

```
                Displacement  Fuel Efficiency
Displacement        1.000000        -0.338914
Fuel Efficiency    -0.338914         1.000000
```

Fig. 16. Correlation Analysis Output

CHAPTER 9 – CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Vehicle Fuel Analysis System demonstrates the use of data-analysis techniques for studying vehicle fuel-efficiency patterns. The Cars India Dataset was loaded and inspected using Python and Pandas. Missing values were treated, duplicate records were checked, and a cleaned dataset was prepared for further analysis.

The project analysed Fuel Efficiency at the vehicle level, compared average fuel efficiency across manufacturers, and studied the relationship between engine Displacement and Fuel Efficiency (correlation −0.3389). The results were presented using scatter plots, bar charts, and histograms.

Overall, the project demonstrates a complete descriptive data-analysis workflow from dataset loading and cleaning to analysis, visualization, and interpretation.

9.2 Limitations

- The analysis is based on the available Cars India Dataset and depends on the coverage and quality of the supplied records.
- The current implementation is primarily descriptive and comparative rather than predictive.
- Traffic conditions, driving behaviour, road conditions, and live sensor measurements are not part of the current dataset.
- The visualizations identify patterns and associations but do not establish causal relationships.

9.3 Future Enhancements

1. Fuel Efficiency Prediction – Develop a machine-learning model to predict fuel efficiency using vehicle specifications such as engine displacement, cylinders, fuel type, and related attributes.

2. GPS Integration – Track vehicle locations, trip routes, and travel patterns, providing additional context for fuel-efficiency analysis.

3. IoT Fuel Monitoring – IoT sensors can collect real-time fuel-consumption information and monitor vehicle performance continuously.

4. Predictive Maintenance – Analyse vehicle-performance indicators to identify possible maintenance requirements at an earlier stage.

5.AI-Based Fuel Consumption Forecasting – Machine-learning or AI-based forecasting to predict fuel-consumption patterns and support fuel planning.

These features are proposed future enhancements and are not part of the current implementation.

CHAPTER 10 – REFERENCES

1. Wes McKinney, Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter, 3rd Edition, O'Reilly Media, 2022.

2. Stefanie Molin, Hands-On Data Analysis with Pandas: A Python Data Science Handbook for Data Collection, Wrangling, Analysis, and Visualization, 2nd Edition, Packt Publishing, 2021.

3. Python Software Foundation, Python Documentation – Python programming language reference and tutorial.

4. pandas Development Team, pandas Documentation – User Guide and API Reference for data loading, cleaning, grouping, and analysis.

5. NumPy Developers, NumPy Documentation – Numerical computing and array operations used in preprocessing.

6. Matplotlib Development Team, Matplotlib Documentation – Visualization and chart generation.

7. Google Research, Google Colaboratory (Colab) – Hosted Jupyter Notebook environment used for project execution.

8. Kaggle, Cars India Dataset – Dataset source referenced in the project documentation.

CHAPTER 11 – APPENDIX – SOURCE CODE

This appendix contains the principal code operations documented in the project materials. The complete source notebook should be attached separately if required by the department.

A. Dataset Loading

```
from google.colab import drive
drive.mount('/content/drive')
import pandas as pd
df = pd.read_csv("Cars India Dataset.csv")
```

B. Dataset Inspection

```
df.head()
df.shape
df.info()
```

C. Missing Value Analysis

```
df.isnull().sum()
df.isnull().sum().sum()      # output: 62
df.select_dtypes(include='number').columns
```

D. Duplicate Check

```
df.duplicated().sum()        # output: 0
```

E. Data Preprocessing

Handling Missing values

```
import numpy as np
numeric_columns = df.select_dtypes(include=np.number).columns
categorical_columns = df.select_dtypes(include='object').columns
```

```
for col in numeric_columns:
    df[col] = df[col].fillna(df[col].median())
for col in categorical_columns:
    df[col] = df[col].fillna(df[col].mode()[0])
df.isnull().sum().sum()      # output: 0

df.duplicated().sum()        # output: 0
```

Handling Duplicates

```
df.drop_duplicates(inplace=True)
print("Duplicates after removal:", df.duplicated().sum())
```

Categorical Encoding

```
df_encoded = pd.get_dummies(df, drop_first=True)
df_encoded.head()
```

Removal of Outliers

```
df[numeric_columns].describe()
df_clean = df.copy()
for col in numeric_columns:
    Q1 = df_clean[col].quantile(0.25)
    Q3 = df_clean[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    df_clean = df_clean[
        (df_clean[col] >= lower) &
        (df_clean[col] <= upper)
    ]
```

Final Check

```
# Final check
print("Final dataset shape:", df_clean.shape)
print("\nMissing values:")
print(df_clean.isnull().sum().sum())
print("\nDuplicate rows:")
print(df_clean.duplicated().sum().sum())
display(df_clean.head())
```

F. Cleaned Dataset Export

```
df_clean.to_csv("Cars_India_Cleaned.csv", index=False)
print("Cleaned dataset saved successfully!")

import pandas as pd
df_clean = pd.read_csv("Cars_India_Cleaned.csv")
print(df_clean.shape)
df_clean.head()
```

G. Fuel Efficiency Comparison

```
fuel_comparison = df_clean[
    ["Model", "Maker", "Fuel Efficiency"]
].sort_values(
    by="Fuel Efficiency",
    ascending=False
)
print(fuel_comparison)
```

H. Manufacturer-wise Analysis

```
manufacturer_analysis = df_clean.groupby("Maker")[
    "Fuel Efficiency"
].mean().sort_values(ascending=False)
print(manufacturer_analysis)
```

I. Engine Size vs Fuel Efficiency

```
# engine displacement vs mileage
engine_mileage = df_clean[
    ["Model", "Displacement", "Fuel Efficiency"]
].sort_values(
    by="Displacement"
)
print(engine_mileage)
```

J. Correlation Analysis

```
# correlation
correlation = df_clean[["Displacement", "Fuel Efficiency"]].corr()
print(correlation)

correlation = df_clean["Displacement"].corr(df_clean["Fuel Efficiency"])
print("Correlation:", correlation)
```

K. Data Visualization

```
import matplotlib.pyplot as plt
# Scatter plot
plt.scatter(df_clean["Displacement"], df_clean["Fuel Efficiency"])
plt.title("Engine Size vs Fuel Efficiency")
plt.xlabel("Engine Displacement (cc)")
```

```
plt.ylabel("Fuel Efficiency (km/L)")
plt.show()

# Bar chart
maker_avg = df_clean.groupby("Maker")["Fuel Efficiency"].mean()
maker_avg = maker_avg.sort_values(ascending=False)
print(maker_avg)
maker_avg.plot(kind="bar", figsize=(8,4))
plt.title("Manufacturer-wise Fuel Efficiency")
plt.xlabel("Manufacturer")
plt.ylabel("Average Mileage (km/L)")
plt.xticks(rotation=45)
plt.show()

# Histogram
plt.hist(df_clean["Fuel Efficiency"], bins=10)
plt.title("Fuel Efficiency Distribution")
plt.xlabel("Fuel Efficiency (km/L)")
plt.ylabel("Number of Vehicles")
plt.show()
```

L. Supplementary – Fuel Type Summary

```
fuel_type_analysis = df_clean.groupby("Fuel")["Fuel Efficiency"].agg(
   ["count", "mean", "min", "max"]
)
print(fuel_type_analysis)
```